# RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

**END SEMESTER EXAMINATION (2022-23) -- SEMESTER 4**

**COURSE: OPERATING SYSTEMS (CS-401)**

Time Allowed: 3 Hours  Maximum Marks: 100 Marks  Paper Code: CS-401

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

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NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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## SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) What is dual-mode operation (User mode vs Kernel mode) in modern operating systems?

Q1. (b) Differentiate between monolithic kernel and microkernel architecture.

Q1. (c) What is Process Control Block (PCB)? List its five essential fields.

Q1. (d) Define turnaround time, waiting time, and response time in CPU scheduling.

Q1. (e) What is Critical Section Problem? List the three required conditions for a valid solution.

Q1. (f) What is race condition? How do mutex locks prevent race conditions?

Q1. (g) State the four necessary conditions for deadlock to occur.

Q1. (h) Define internal and external fragmentation in memory management.

Q1. (i) What is thrashing? How does working set model prevent thrashing?

Q1. (j) Explain Belady Anomaly in FIFO page replacement algorithm.

## SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Given processes P1, P2, P3, P4 with arrival times (0, 1, 2, 3) and burst times (8, 4, 9, 5). Draw Gantt charts for FCFS, SJF (Pree

Q2. (b) Solve Banker Deadlock Avoidance algorithm: Determine whether the system is in safe state, find safe sequence, and check if re

Q2. (c) For memory reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 with 3 frames, calculate page faults for FIFO, LR

Q2. (d) Given disk requests: 98, 183, 37, 122, 14, 124, 65, 67 with initial head at 53. Calculate total seek time for FCFS, SSTF, SCAN, a

Q2. (e) Explain Peterson algorithm for two-process mutual exclusion and prove mutual exclusion, progress, and bounded waiting.

## SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

**Q3. (a) Explain Dining Philosophers problem and write solution using counting semaphores without deadlock. [10 Marks]**

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(b) Explain Readers-Writers problem with semaphore solution prioritizing readers. [10 Marks]

**Q4. (a) Explain multi-level feedback queue scheduling with priority aging and preemption policies. [10 Marks]**

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(b) Explain thread models (Many-to-One, One-to-One, Many-to-Many) and POSIX pthread library. [10 Marks]

**Q5. (a) Explain paging hardware with Translation Lookaside Buffer (TLB) and calculate effective access time. [10 Marks]**

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(b) Explain inverted page tables, multi-level hierarchical paging, and segmentation with protection. [10 Marks]

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**Q6. (a) Explain Deadlock Detection and Recovery techniques using Resource Allocation Graph (RAG). [10 Marks]**

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(b) Explain deadlock prevention by eliminating hold-and-wait, mutual exclusion, and circular wait. [10 Marks]

**Q7. (a) Explain File Allocation methods: Contiguous, Linked, and Indexed allocation with diagrams. [10 Marks]**

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(b) Explain Linux EXT4 file system inode structure, directory management, and free space management. [10 Marks]

## EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

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***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****